



Using fear, greed and machine learning for optimizing global portfolios: A Black-Litterman approach

Ronil Barua, Anil K. Sharma ^{*}

Department of Management Studies, Indian Institute of Technology Roorkee, Uttarakhand 247667, India

ARTICLE INFO

JEL classification:

C45
C53
G11
G41

Keywords:

Black-Litterman
Investor sentiment
Financial forecasting
CEEMDAN
GRU
XGBoost

ABSTRACT

We introduce a new dimension in constructing relative investor views for the Black-Litterman model by incorporating fear/greed technical indicator predictions as a proxy for investor sentiment in the portfolio construction process. We apply a hybrid CEEMDAN-GRU deep learning model to forecast this indicator and the XGBoost ensemble learning algorithm to forecast returns for ten country ETFs and create relative views for the Black-Litterman model. These models beat several benchmark forecasting models. Our empirical results show that the proposed approach outperforms the Markowitz, minimum-variance, equally-weighted and risk-parity strategies along with four other Black-Litterman approaches from the literature for six investment periods.

1. Introduction

The Black-Litterman (BL) model was created to overcome several limitations of the Markowitz approach, such as corner solutions, model estimation error, and unintuitive portfolios by allowing investors to integrate subjective 'views' in the form of future expected returns with market-implied returns using Bayesian principles to arrive at an allocation based on the updated return distribution (Black and Litterman, 1990, 1992). These views on assets can be 'absolute' such as "X will give a return of 20%," or 'relative' such as "X will surpass Y by 10%". New methods and procedures for deriving investor views for the BL model is an avenue which needs more research (Fernandes et al., 2012; Silva et al., 2017). Machine learning models have advantages in forecasting due to their high-dimensional nature, and they have gained significant attention in creating BL views (Mullainathan and Spiess, 2017; Kolm et al., 2021). Creamer (2015) performed machine learning based forecasting using news sentiment, corporate social indicators, and analysts' recommendations to design views for the BL model. Kara et al. (2019) used a hybrid model combining GARCH and support vector regression to forecast stock returns and generate BL views. Rezaei et al. (2021) used a hybrid CEEMD-CNN-LSTM deep learning model for constructing BL views in the form of return forecasts which resulted in portfolios with higher returns and less concentrated allocations. Mourtas and Katsikis (2022) used a three-layer feedforward neural network (MAWTSNN) for creating BL views. Barua and Sharma (2022) used a hybrid CNN-BiLSTM deep learning prediction model for constructing BL views, with the resulting portfolio outperforming three other benchmark strategies. The aforementioned studies, although very impressive in findings, have two important issues: (a) They focus on mainly constructing absolute BL views and do not provide any clear mechanism for specifying

^{*} Corresponding author.

E-mail addresses: rbarua@bm.iitr.ac.in (R. Barua), anilsharma@ms.iitr.ac.in (A.K. Sharma).

Table 1
Ten iShares MSCI country ETFs used.

1.	iShares MSCI Australia ^d
2.	iShares MSCI Brazil ^e
3.	iShares MSCI Canada ^d
4.	iShares MSCI Germany ^d
5.	iShares MSCI Japan ^d
6.	iShares MSCI Mexico ^e
7.	iShares MSCI South Africa ^e
8.	iShares MSCI South Korea ^e
9.	iShares MSCI Taiwan ^e
10.	iShares MSCI United Kingdom ^d

Note: ^d and ^e imply whether the economy is developed or emerging, respectively as per IMF classification.

relative views which are better substitutes for investor preferences and more closely represent the way investment managers feel about different assets; (b) They do not compare the efficacy of the machine learning based BL portfolios with BL models in the prior literature which used traditional econometric techniques for generating views in order to truly uncover the benefits of machine learning based forecasting in portfolio optimization. We intend to address these issues in this letter.

The sentiment of investors needs to be captured while formulating subjective BL views. Technical indicators contain information about the psychological state of the market and are a valuable tool for analyzing investor sentiment (Shefrin, 2002; Fang et al., 2014; Khayamim et al., 2018). Several researchers have used different technical indicators as proxies for measuring investor sentiment, such as relative strength index, psychological line index, adjusted turnover rate, and trading volume (Yang and Zhou, 2015, 2016; Gao and Yang, 2017). Buy and sell imbalance is a parsimonious measure of investor sentiment (Yang and Gao, 2014). One such indicator to measure this imbalance is the fear/greed (FG) indicator which is the ratio of buying strength (bulls) to selling strength (bears) at a particular time. The higher (lesser) the reading of this indicator, investors are greedier (fearful), which tends to move prices up (down). This indicator is derived from an amalgamation of various factors such as put/call ratio, volatility, advance/decline ratio and more. The FG indicator can provide substantial economic value if used by investors to time the market (Day et al., 2019). Based on this analogy, we capture the psychological component of the market by employing the signals emitted by the FG indicator predictions as a tool to categorize assets upon which we express relative BL views, which is an entirely novel approach. We use a two-step procedure for predicting the FG indicator. First, technical indicators carry a lot of noise, and decomposing them into frequency-specific components can improve predictability (Stein, 2022). Abiding by this, we apply a denoising technique called complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) on the FG indicator. CEEMDAN has shown significant improvement in predictive accuracy by eliminating noise from financial time series (Cao et al., 2019; Lin et al., 2022). Second, neural networks are more capable of handling non-linear and non-stationary signals as they contain multiple layers for processing and can better identify features of any financial time series compared to traditional econometric models (Krauss et al., 2017). Recurrent neural networks (RNNs) are among the most used and effective in financial forecasting, which contain feedback loops and have consistently delivered superior performance (Fischer and Krauss, 2018; Ma et al., 2023). We employ a variant of RNNs called gated recurrent unit (GRU) for forecasting the FG indicator after decomposition via CEEMDAN. The CEEMDAN-GRU hybrid model has been previously used by Wang et al. (2021) and Feng et al. (2022) for forecasting natural gas prices and solar output levels, respectively. We compare the performance of this hybrid model with two commonly used methods for financial forecasting, namely auto-regressive integrated moving average (ARIMA) and multi-layer perceptron neural network (MLP).

Along with FG indicator prediction, we also require assets' return prediction to determine the magnitude of outperformance among assets when building relative BL views. Stock returns follow a very non-linear, evolutionary, and complex nature, and researchers have demonstrated that machine learning algorithms can better model the relationship between returns and predictor variables, which can lead to significantly better predictions compared to traditional time series models (Gu et al., 2020; Leippold et al., 2022). Technical indicators possess tremendous potential for stock return prediction when used as predictor variables (Neely et al., 2014; Alhashel et al., 2018). Ensemble machine learning models have been shown to deliver the best performance in return prediction across time due to their ability to integrate several weak learners into a strong one, thereby minimizing overfitting and overcoming the bias-variance trade-off (Hanauer and Kalsbach, 2023; Huddleston et al., 2023). One such ensemble technique is extreme gradient boosting (XGBoost) which has been widely used in classification and regression tasks (Abbasimehr et al., 2023; Dezhkam and Manzuri, 2023). We use XGBoost for forecasting returns with moving average and momentum indicators as predictors and compare its performance with four other regression-based machine learning algorithms used in the prior literature for stock return forecasting, viz. stochastic gradient descent (SGD) regression (Gupta and Chatterjee, 2021), partial least squares (PLS) regression (Cengiz and Herwartz, 2011), least absolute shrinkage and selection operator (LASSO) regression (Ma et al., 2022) and elastic-net (E-Net) regression (Wu et al., 2019).

After building the views, we consider six investment periods and find that the BL portfolios created by our proposed procedure outperform the Markowitz, minimum-variance, risk-parity, and equally-weighted portfolio strategies along with four other BL approaches from the literature based on traditional models for generating views viz. BL with views derived via predictive regression (Lin et al., 2021), BL with sample mean returns as views (Bessler et al., 2017, 2021), BL with views generated from a momentum strategy (Fabozzi et al., 2006; Tang et al., 2021) and BL with EGARCH-in-mean derived views (Beach and Orlov, 2007; Duqi et al., 2014).

We describe the data used in Section 2 and explain the methodology adopted in Section 3. Experimental results are presented in

Table 2
Technical Indicators used.

1.	Moving Average (1,6)
2.	Moving Average (1,9)
3.	Moving Average (1,12)
4.	Moving Average (2,6)
5.	Moving Average (2,9)
6.	Moving Average (2,12)
7.	Moving Average (3,6)
8.	Moving Average (3,9)
9.	Moving Average (3,12)
10.	Momentum (1)
11.	Momentum (3)
12.	Momentum (6)
13.	Momentum (9)
14.	Momentum (12).

Section 4, and Section 5 contains the concluding remarks.

2. Data and explanation

We focus on building a diverse international portfolio comprising five developed market ETFs and five emerging market ETFs to test our approach's universal applicability. For this purpose, we collect daily price, volume, and fear/greed indicator data on ten iShares MSCI country ETFs for the period 1/1/2008 - 12/31/2022, the details of which are provided in Table 1. We additionally collect daily data on several moving average and momentum indicator combinations for all ETFs for the same period, as shown in Table 2, to be used as predictors for return forecasting in line with prior studies (Stein, 2022; Fernández et al., 2023). We use the US 3-month treasury bill rate as the risk-free rate and the MSCI World index as the market index. All data is acquired from the Bloomberg terminal. The fear/greed indicator provided by Bloomberg is an oscillator which is based on the on-balance computation of the true range and is widely used for divergence analysis and recognizing trend persistence. It is the ratio of buying strength to selling strength and the speed of the oscillator is controlled by three parameters which determine the periods used by the two weighted moving averages applied to the on-balance true range:

"First LookBack" specifies the first weighted moving average period.

For the second moving average period:

- If "Look Back Period" is not 0, then it is used directly.
- If "Look Back Period" is 0, the second period is the truncated value of the Sensitivity squared.

3. Methodology

3.1. Black-Litterman (BL) model

Walters (2014) provides a detailed description of the BL model. The BL model integrates inputs from two sources: the subjective return estimates on assets (investors' views) and implied returns from the equilibrium market portfolio. The prior distribution of returns is given as:

$$\mu_{prior} \sim N(\Pi, \tau \Sigma) \quad (1)$$

where, Π denotes the implied returns, τ (tau) is a scalar that acts as a 'weight-on-views' parameter and represents the uncertainty of equilibrium returns. Smaller values of τ place more weight on the market equilibrium returns, and larger values give more weight to the investor's views, thereby allowing them to have a stronger influence on the expected returns. Σ is the covariance matrix. Implied returns can be derived by reverse optimization and is given as:

$$\Pi = \lambda \sum \omega_{mkt} \quad (2)$$

where, λ is the risk-aversion coefficient and ω_{mkt} is the vector of benchmark portfolio weights. We obtain the risk-aversion coefficient as:

$$\lambda = (r_{mkt} - r_f) / \sigma^2 \quad (3)$$

where, r_{mkt} is the market expected return, r_f is the risk-free rate, and σ^2 is the variance of the market portfolio. The posterior return estimates (μ_{BL}) are given as:

$$\mu_{BL} = \left[\left(\tau \sum \right)^{-1} + P^T \Omega^{-1} P \right]^{-1} \left[\left(\tau \sum \right)^{-1} \pi + P^T \Omega^{-1} Q \right] \quad (4)$$

The posterior covariance matrix ($\sum_{BL}$) is given as:

$$\sum_{BL} = \sum + \left[\left(\tau \sum \right)^{-1} + P^T \Omega^{-1} P \right]^{-1} \quad (5)$$

Throughout this section, K denotes the number of views, and N is used to indicate the number of assets. P is a K X N matrix with each row specifying a view. For an investor having 'x' views on 'y' assets, the matrix is given as:

$$P = \begin{bmatrix} p_{11} & \cdots & p_{1y} \\ \vdots & \ddots & \vdots \\ p_{x1} & \cdots & p_{xy} \end{bmatrix} \quad (6)$$

where, $\sum_{t=1}^y P_{at} = 1$, for an absolute view 'a' and $\sum_{t=1}^y P_{rt} = 0$, for a relative view 'r'.

Q is a K X 1 matrix containing each view's subjective return estimate. We initially choose τ as 0.10 and then perform sensitivity analysis by calibrating values between 0.025 to 1, which acts as a test of robustness and has been the most reliable and followed approach in the prior BL literature (Beach and Orlov, 2007; Creamer, 2015; Bessler et al., 2017; Pyo and Lee, 2018; Han and Li, 2023). The uncertainty of the views forms a normally distributed error term vector (ε) given as: $\varepsilon \sim N(0, \Omega)$. The variances of the error terms (ω) represent the uncertainty of the views and form Ω , where Ω represents a K X K diagonal covariance matrix with zeros in all its off-diagonal positions as the model assumes that the views are independent of one another. We follow He and Litterman (2002) and compute Ω as:

$$\Omega = \tau * P \sum P^T \quad (7)$$

As a further test of robustness, we construct an alternative formulation of the BL portfolios using Idrozek's (2007) procedure, which proposes a different technique for figuring out the implied confidence levels in the views and demonstrates how an implied confidence level structure can be integrated with an instinctive 0% to 100% confidence level specified by the investor in each view to determine the values of Ω , while at the same time eliminating the challenge of specifying a value for τ . In the entire process, τ is held constant and does not affect μ_{BL} , which removes the difficulties associated with its specification, thereby simplifying the investor's process of defining the uncertainty in the views. Simply put, the value of τ is irrelevant in this approach as it cancels out in the matrix multiplications. The diagonal elements of Ω are calculated based on the investor-specified confidence levels, resulting in portfolio tilts approximately equal to the difference between the weight vector based on 100% confidence and the market capitalization weight vector ($w_{100\%} - w_{mkt}$) multiplied by the confidence level (C) specified by the investor:

$$Tilt_k \approx (w_{100\%} - w_{mkt}) * C_k \quad (8)$$

where, $Tilt_k$ is the approximate tilt caused by the kth view (N x 1 column vector), and C_k is the confidence in the kth view. If there are no other views, the approximate recommended weight vector resulting from the view is given as:

$$w_{k,\%} \approx w_{mkt} + Tilt_k \quad (9)$$

where, $w_{k,\%}$ is the target weight vector based on the tilt caused by the kth view (N x 1 column vector).

3.2. Complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN)

Empirical model decomposition (EMD) is an effective technique that can separate noise from any financial time series by decomposing the series into several intrinsic mode functions (IMFs), each representing a specific characteristic of the original signal at different time scales. However, EMD suffers from the issues of 'mode mixing' which occurs when the IMFs are not well-separated leading to overfitting and reconstruction errors (Lu and Ma, 2020). To overcome these issues, Torres et al. (2011) proposed the CEEMDAN algorithm, which eliminates mode mixing and exhibits negligible reconstruction errors by introducing the concept of adaptive noise to the input signal and then decomposes the noisy signal into IMFs multiple times to create multiple realizations of IMFs. The steps of CEEMDAN followed in this letter for decomposing the FG indicator series are demonstrated below:

Step I: We add a Gaussian white noise sequence $w_i(t)$ with standard deviation ε (used to set the signal-to-noise ratio for each decomposition stage) to the original FG indicator series $FG(t)$ given as:

$$FG^i(t) = FG(t) + \varepsilon_0 w_i(t), i = 1, 2, 3, \dots, I \quad (10)$$

where, I represents the number of realizations.

Step II: We perform EMD on $FG^i(t)$ after each realization to obtain the first IMF, $\overline{IMF}_1(t)$:

$$\overline{IMF}_1(t) = \frac{1}{I} \sum_{i=1}^I IMF_1^i(t) \quad (11)$$

The residual, $r_1(t)$ of this very first stage is given by:

$$r_1(t) = FG(t) - \overline{IMF}_1(t) \quad (12)$$

Step III: We define $E_j(\cdot)$ as an operator which gives the j^{th} IMF obtained by EMD. We obtain the second IMF as:

$$\overline{IMF}_2(t) = \frac{1}{I} \sum_{i=1}^I E_1\{r_1(t) + \varepsilon_1 E_1[w_i(t)]\} \quad (13)$$

The corresponding residual is given as:

$$r_2(t) = r_1(t) - \overline{IMF}_2(t) \quad (14)$$

Step IV: For the following stages, we calculate the k^{th} residue as:

$$r_k(t) = r_{k-1}(t) - \overline{IMF}_k(t) \quad (15)$$

The $(k + 1)^{\text{th}}$ IMF can be obtained as:

$$\overline{IMF}_{k+1}(t) = \frac{1}{I} \sum_{i=1}^I E_1\{r_k(t) + \varepsilon_k E_k[w_i(t)]\} \quad (16)$$

Step V: We repeat step IV until the residue component is no longer viable to be decomposed, thereby making no more extraction of IMFs possible, and a total of K IMFs are obtained. Finally, the original FG indicator series can be decomposed as the summation of IMFs and the final residue, $R(t)$ given as:

$$FG(t) = \sum_{k=1}^K \overline{IMF}_k(t) + R(t) \quad (17)$$

3.3. Gated recurrent unit (GRU)

GRU neural network is a variant of RNNs introduced by [Choi et al. \(2014\)](#). Compared with other RNN architectures such as long short-term memory (LSTM), GRU virtually maintains the same accuracy with lesser model parameters, is faster to train, possesses an update gate for dynamically adjusting the information flow and a reset gate for deciding on whether the previous hidden state should be considered or ignored. The following equations explain the functioning of GRU:

$$u_t = \sigma(w_u \cdot [c_{t-1}, x_t]) \quad (18)$$

$$r_t = \sigma(w_r \cdot [c_{t-1}, x_t]) \quad (19)$$

where, σ is the sigmoid function, c_{t-1} denotes the previous hidden state, x_t denotes present input, u_t is the update gate and r_t is the reset gate. w_u and w_r are the weight matrices of the update gate and reset gate respectively.

The candidate hidden state $\tilde{c}_t$ is given as:

$$\tilde{c}_t = \tanh(w \cdot [r_t \odot c_{t-1}, x_t]) \quad (20)$$

where, w represents the weight matrix of the reset gate and $\odot$ denotes element-wise multiplication. The final output of the current time step is given by the following equation where c_t denotes the hidden state vector:

$$c_t = u_t \odot \tilde{c}_t + (1 - u_t) \odot c_{t-1} \quad (21)$$

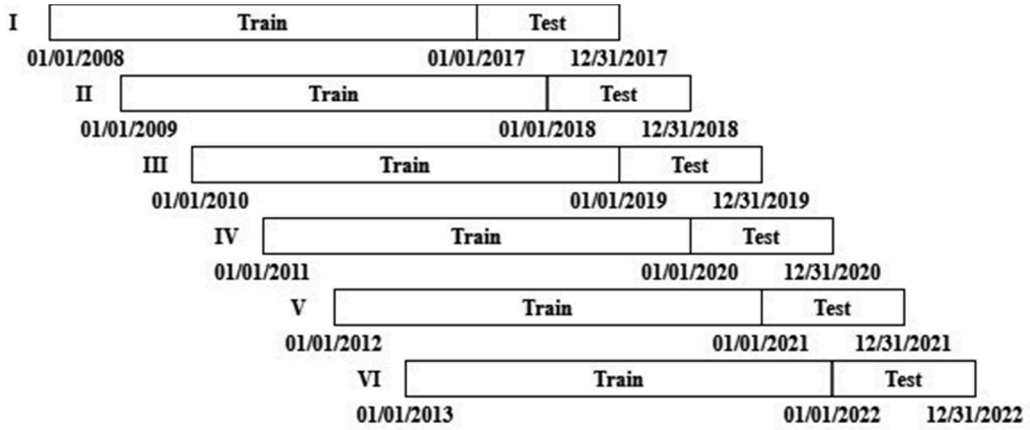


Fig. 1. Train-test set distribution into six periods.

3.4. Extreme gradient boosting (XGBoost)

XGBoost is an ensemble machine learning algorithm based on gradient boosting trees introduced by [Chen and Guestrin \(2016\)](#). XGBoost follows an additive process by integrating multiple weak learners into a strong learner and uses K regression trees for making predictions:

$$\hat{p}_i = \sum_{k=1}^K f_k(x_i), f_k \in F \quad (22)$$

where, $\hat{p}_i$ is the predicted value, f represents a tree in tree space F , K denotes the number of regression trees and x_i denotes the i^{th} instance. XGBoost has an effective way to tackle overfitting by comprising of a regularized model. The objective function to be optimized is given as:

$$L(\varphi) = \sum_i l(\hat{p}_i, p_i) + \sum_k \Omega(f_k) \quad (23)$$

where, l is the loss function, Ω is the penalty term and $\Omega(f) = \gamma T + 0.5\lambda \|w\|^2$, where, γ and λ are parameters controlling the complexity of the tree, T is the number of leaves in the tree and w denotes the tree weight for internal split. By consulting prior literature, we set the number of trees as 300 and keep the rest of the hyper-parameters as default.

3.5. Prediction evaluation parameters

We use the following two metrics to determine the accuracy of the FG indicator and return prediction models, which are among the most used in the financial prediction literature. The smallest values of these metrics denote the most accurate predictions among the models.

$$\text{Mean Absolute Error (MAE)} = \frac{1}{n} \sum_{i=1}^n |p_i - a_i| \quad (24)$$

$$\text{Root Mean Squared Error (RMSE)} = \sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - a_i)^2} \quad (25)$$

where, n denotes total observations, p_i denotes the predicted value, and a_i denotes the actual value.

4. Experimental results

We use Python 3.9 along with TensorFlow, scikit-learn, PyEMD, XGBoost, and PyPortfolioOpt libraries for our computations. For the prediction tasks, we subdivide our dataset into six periods with each period containing ten years of data in which 90% is used as the training set and the remaining 10% as the test set. [Fig. 1](#) highlights this split.

Table 3
Hyperparameter tuning for GRU.

Hyperparameter	Range of Values	Chosen Value
Number of hidden layers	[1, 3, 5]	3
Number of neurons in the layers	[25, 50, 75, 125, 150, 200]	150, 75, 50
Dropout rate	[0.01, 0.1, 0.3]	0.1
Epochs	[50, 100, 200]	100
Batch size	[32, 64, 128]	64
Activation function	[tanh, Linear, ReLU]	ReLU
Optimizer	[Adam, SGD, RMSprop]	Adam

Table 4
Error metrics for fear/greed indicator prediction of ETFs on the test sets.

		ARIMA		MLP		GRU		CEEMDAN-GRU	
		RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
2017	Minimum	0.429	0.347	0.152	0.125	0.121	0.094	0.055	0.044
	Maximum	2.061	1.657	1.407	1.241	0.674	0.545	0.291	0.233
	Average	1.027	0.842	0.454	0.382	0.289	0.239	0.160	0.127
	Standard Deviation	0.582	0.468	0.406	0.355	0.182	0.152	0.104	0.082
2018	Minimum	0.573	0.431	0.167	0.132	0.151	0.131	0.092	0.069
	Maximum	2.830	2.314	1.293	1.042	0.908	0.793	0.546	0.436
	Average	1.526	1.221	0.529	0.425	0.415	0.353	0.228	0.177
	Standard Deviation	0.763	0.638	0.353	0.287	0.255	0.217	0.141	0.113
2019	Minimum	0.490	0.385	0.120	0.097	0.110	0.093	0.073	0.056
	Maximum	2.599	2.043	0.660	0.545	0.538	0.466	0.342	0.271
	Average	1.279	1.038	0.339	0.276	0.284	0.246	0.161	0.128
	Standard Deviation	0.697	0.561	0.190	0.157	0.154	0.137	0.094	0.076
2020	Minimum	1.331	0.892	0.333	0.241	0.284	0.215	0.145	0.098
	Maximum	3.469	2.578	1.004	0.764	0.784	0.630	0.379	0.285
	Average	2.348	1.613	0.624	0.458	0.508	0.396	0.255	0.189
	Standard Deviation	0.823	0.605	0.230	0.184	0.167	0.142	0.093	0.075
2021	Minimum	0.578	0.474	0.182	0.139	0.175	0.143	0.087	0.071
	Maximum	2.549	1.993	0.766	0.632	0.710	0.608	0.396	0.316
	Average	1.358	1.078	0.498	0.396	0.387	0.330	0.207	0.163
	Standard Deviation	0.629	0.490	0.214	0.176	0.171	0.153	0.099	0.079
2022	Minimum	1.159	0.919	0.304	0.238	0.243	0.207	0.118	0.094
	Maximum	2.756	2.222	0.859	0.672	0.663	0.589	0.371	0.281
	Average	2.021	1.621	0.579	0.467	0.452	0.390	0.251	0.196
	Standard Deviation	0.583	0.460	0.177	0.148	0.137	0.124	0.089	0.070

Note: Due to space limitations, we do not report the error metrics for all 10 ETFs individually for six time periods. Instead, we show the minimum, maximum, average and standard deviation of the errors under each model. **Legend:** ARIMA = auto-regressive integrated moving average; MLP = multi-layer perceptron; GRU = gated recurrent unit; CEEMDAN = complete ensemble empirical mode decomposition with adaptive noise.

4.1. Fear/greed indicator prediction

The FG indicator series is normalized in the range of 0 to 1 before training by neural networks. For determining the optimal hyperparameters for the GRU neural network, we apply the grid search algorithm to find the combination achieving the best model performance. The different combinations specified have been selected based on the prior literature, and the final combination chosen is highlighted in [Table 3](#). Additionally, we select 'Huber' as the loss function to deal with outliers and five days as the lookback period. [Table 4](#) reports the one-step-ahead FG indicator prediction metrics for all models for the six test sets as per [Fig. 1](#). We observe that GRU achieves the lowest average RMSE and MAE values and the lowest standard deviation for all six periods compared to ARIMA, an exemplary traditional econometric model, and MLP, a feedforward deep learning model. The benefits of CEEMDAN are well substantiated by the fact that compared to GRU, CEEMDAN-GRU achieves a reduction in average RMSE of 44.64% (2017), 45.06% (2018), 43.31% (2019), 49.80% (2020), 46.51% (2021) and 44.47% (2022). This result proves that combining denoising with neural networks can be a very efficient technique for predicting complicated financial time series. Overall, our CEEMDAN-GRU hybrid model gives the most accurate predictions for all six periods, highlighting its robustness.

4.2. Return prediction

[Table 5](#) shows the one-step-ahead daily logarithmic return prediction results for all models for each of the six test sets as per [Fig. 1](#). The technical indicators highlighted in [Table 2](#) for each ETF are used as predictors to predict daily returns. Before training, the indicators are standardized in the range of -1 to 1 as a part of feature scaling. XGBoost stands robust and outperforms all other models by achieving the lowest average RMSE and MSE values and the lowest standard deviation for all six periods. This finding contributes to the empirical literature that ensemble machine learning models are superior to their regression-based counterparts for financial return

Table 5

Error metrics for return prediction of ETFs on the test sets.

		SGD		PLS		LASSO		E-Net		XGBoost	
		RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
		($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)	($\times 10^{-3}$)
2017	Minimum	0.476	0.360	2.318	1.840	4.686	3.537	2.668	2.012	0.187	0.119
	Maximum	6.107	4.553	9.709	6.852	19.594	13.155	14.918	9.944	2.728	0.837
	Average	2.490	1.778	5.682	4.548	9.143	6.759	5.783	4.241	0.728	0.361
	Standard Deviation	2.055	1.408	2.615	2.057	4.520	3.129	3.644	2.458	0.749	0.226
2018	Minimum	0.785	0.577	3.777	2.893	9.057	6.847	5.617	4.091	0.342	0.189
	Maximum	6.003	4.439	9.797	7.578	22.864	17.896	15.712	12.535	2.588	0.881
	Average	3.189	2.291	6.571	5.165	13.743	10.606	8.947	6.871	1.253	0.542
	Standard Deviation	1.876	1.444	2.221	1.777	4.858	3.953	3.959	3.174	0.747	0.281
2019	Minimum	0.381	0.258	2.530	1.924	6.043	4.455	4.072	2.984	0.193	0.089
	Maximum	2.444	1.854	7.933	6.409	16.266	12.660	11.572	8.978	1.025	0.605
	Average	1.386	1.000	4.624	3.683	10.015	7.618	6.839	5.184	0.458	0.255
	Standard Deviation	0.845	0.638	1.712	1.425	3.357	2.682	2.596	2.062	0.272	0.158
2020	Minimum	2.484	1.680	5.862	4.050	16.197	10.448	11.567	7.271	3.051	0.882
	Maximum	16.163	9.021	22.059	15.333	40.920	26.252	34.909	22.114	13.612	4.758
	Average	7.774	4.386	12.898	8.475	26.207	16.786	21.218	13.327	7.727	2.558
	Standard Deviation	4.548	2.702	5.348	3.655	6.951	4.528	6.699	4.325	3.418	1.386
2021	Minimum	0.645	0.510	3.566	2.693	8.931	6.966	5.102	3.967	0.603	0.261
	Maximum	11.497	8.418	12.040	10.766	20.460	15.799	14.084	10.827	5.931	2.932
	Average	3.950	2.970	7.734	6.327	12.451	9.638	7.674	5.920	2.003	0.993
	Standard Deviation	3.241	2.421	3.011	2.571	3.751	2.971	3.028	2.396	1.904	0.865
2022	Minimum	1.014	0.743	4.725	3.702	12.506	9.891	8.370	6.600	0.584	0.342
	Maximum	6.818	4.953	9.865	7.693	23.054	17.391	16.253	12.220	6.255	1.840
	Average	2.977	2.141	7.330	5.671	17.135	13.002	11.916	8.978	2.362	0.835
	Standard Deviation	1.820	1.382	1.925	1.496	3.372	2.374	2.770	1.944	2.052	0.482

Note: Due to space limitations, we do not report the error metrics for all 10 ETFs individually for six time periods. Instead, we show the minimum, maximum, average and standard deviation of the errors under each model. **Legend:** SGD = stochastic gradient descent; PLS = partial least squares; LASSO = least absolute shrinkage and selection operator; E-Net = elastic-net; XGBoost = extreme gradient boosting.

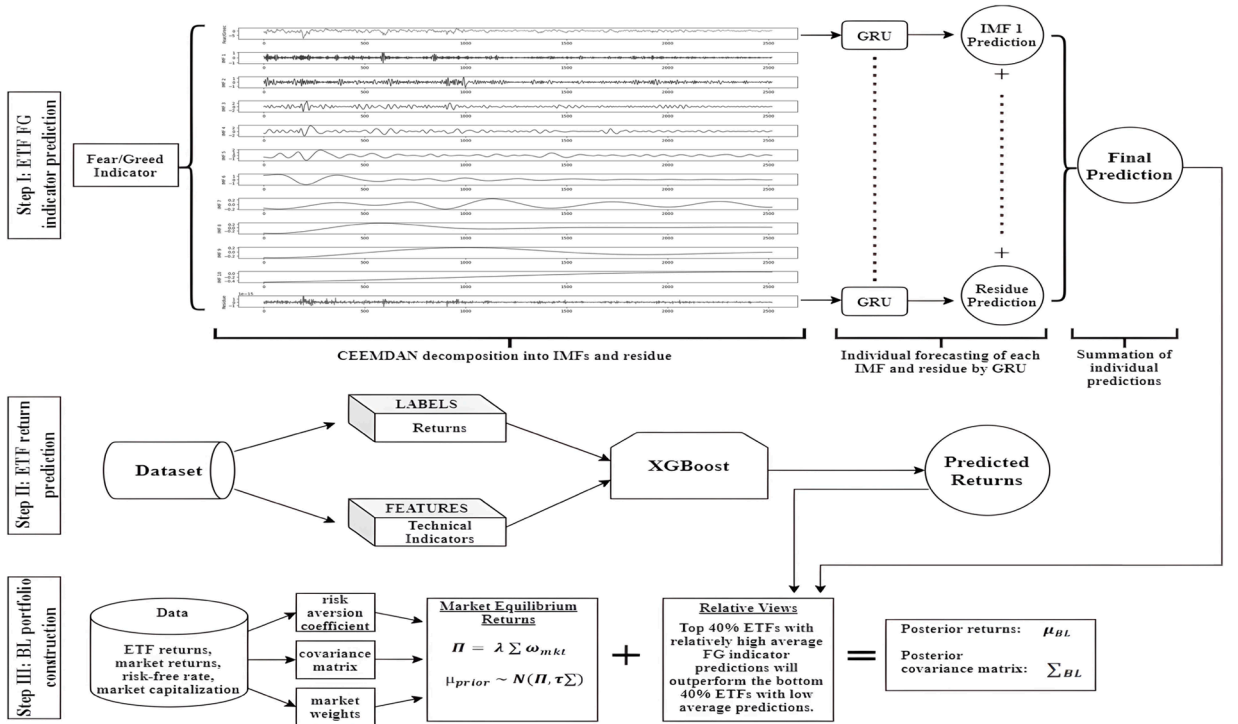
**Fig. 2.** Empirical design of the proposed Black-Litterman procedure.

Table 6

Performance evaluation of portfolios.

		BL-FG (Standard)	BL-FG (Idzorek)	BL-PR	BL- EM	BL-SM	BL- Mom	Mean- Var	Min- Var	1/N	RP
2017	Expected Return	27.1%	26.2%	19.1%	12.6%	8.5%	26.1%	16.1%	10.8%	22.6%	18.8%
	Volatility	7.5%	7.2%	7.4%	8.3%	10.00%	15.1%	7.2%	7.4%	8.7%	7.1%
	Sharpe Ratio	3.46	3.52	2.47	1.41	0.75	1.66	2.09	1.34	2.50	2.49
	Sortino Ratio	4.66	4.50	3.54	1.71	0.68	2.11	2.70	1.88	3.88	2.85
2018	Expected Return	12.0%	12.7%	-8.1%	4.6%	5.00%	6.3%	-10.9%	-17.1%	-16.22%	-20.0%
	Volatility	20.8%	22.2%	11.2%	11.9%	13.3%	13.8%	17.0%	10.9%	13.65%	12.0%
	Sharpe Ratio	0.48	0.48	-0.90	0.22	0.23	0.31	-0.75	-1.76	-1.33	-1.82
	Sortino Ratio	0.60	0.61	-1.12	0.24	0.23	0.37	-0.97	-2.03	-1.69	-2.26
2019	Expected Return	23.4%	27.6%	19.7%	9.7%	17.6%	16.0%	18.8%	12.0%	16.09%	15.3%
	Volatility	10.1%	9.5%	9.4%	11.6%	22.0%	13.0%	9.9%	8.7%	10.45%	9.5%
	Sharpe Ratio	2.11	2.68	1.86	0.66	0.71	1.06	1.69	1.14	1.34	1.39
	Sortino Ratio	2.51	3.00	2.40	0.85	0.99	1.38	1.61	1.35	1.91	1.60
2020	Expected Return	17.8%	18.7%	6.8%	5.2%	6.9%	9.5%	10.3%	4.2%	9.74%	8.5%
	Volatility	25.7%	23.8%	22.2%	21.8%	26.5%	30.4%	25.3%	18.1%	27.46%	24.9%
	Sharpe Ratio	0.68	0.77	0.29	0.22	0.25	0.30	0.39	0.21	0.34	0.33
	Sortino Ratio	0.66	0.81	0.34	0.24	0.27	0.37	0.47	0.22	0.45	0.43
2021	Expected Return	29.2%	38.1%	15.9%	8.9%	10.2%	16.3%	16.4%	9.2%	5.76%	4.1%
	Volatility	12.7%	12.9%	11.5%	12.6%	17.3%	16.5%	11.9%	10.4%	12.23%	11.1%
	Sharpe Ratio	2.29	2.95	1.38	0.70	0.59	0.98	1.38	0.88	0.47	0.36
	Sortino Ratio	3.13	3.70	1.74	0.84	0.57	1.25	1.49	1.14	0.64	0.54
2022	Expected Return	14.6%	14.2%	-19.3%	5.0%	10.6%	13.0%	-1.1%	-13.5%	-13.77%	-15.4%
	Volatility	20.4%	19.6%	18.3%	16.8%	20.5%	22.2%	20.7%	16.0%	18.28%	17.2%
	Sharpe Ratio	0.62	0.62	-1.17	0.18	0.42	0.49	-0.15	-0.97	-0.86	-1.01
	Sortino Ratio	0.73	0.67	-1.64	0.21	0.51	0.63	-0.21	-1.18	-1.17	-1.22

Note: This table reports portfolio performance measures for six evaluation periods which are the test sets from Fig. 1. τ is considered as 0.10 for all BL models. **Legend:** BL-FG = proposed BL approach; BL-PR = BL with views derived via predictive regression; BL-EM = BL with EGARCH-in-mean derived views; BL-SM = BL with sample mean returns as views; BL-Mom = BL with views generated from a momentum strategy; Mean-Var = Markowitz portfolio; Min-Var = minimum variance portfolio; 1/N = equally-weighted portfolio; RP = risk-parity portfolio.

prediction, especially when using technical indicators as predictors.

4.3. Portfolio construction and evaluation

We consider the six test sets from Fig. 1 as investment periods of one year each, as most investors have medium to long horizons and follow buy-and-hold strategies (Oikonomou et al., 2018). We perform portfolio optimization with our asset universe of 10 ETFs. Fig. 2 showcases our BL procedure. We use the FG indicator predictions from the CEEMDAN-GRU model for each ETF as a technical strategy to create relative BL views in which the top 40% ETFs with relatively bullish sentiment outperform the bottom 40% bearish ones by a magnitude that is determined by the returns predicted via the XGBoost model. We ignore short-selling and consider transaction costs as 50 basis points. We employ the Sharpe and Sortino ratios as portfolio evaluation measures. Table 6 displays the performance evaluation of portfolios for each of the six investment periods. Our model, BL-FG (Standard), yields the highest Sharpe and Sortino ratios for each of the six periods compared to all eight benchmark strategies. An alternate formulation of our model using Idzorek's process is labeled BL FG (Idzorek). In our case, since the views are derived from a prediction model, the confidence levels for the views in the BL FG (Idzorek) portfolios come from the out-of-sample R^2 of the predicted returns via the XGBoost model. To be more precise, we found it imperative that for each relative view, the average of the R^2_{OOS} of the predicted returns of the involved assets be taken as the confidence for that view. As the results show, this alternative formulation also outperforms all benchmark strategies. Markets struggled globally in 2018, primarily due to the US-China trade tensions, and 2022 experienced a global economic slowdown triggered by the Russia-Ukraine war, leading to high inflation and geopolitical disquietude. Most portfolio strategies deliver negative performance for 2018 and 2022; however, our proposed approach achieves positive performance even during these two years of turmoil. We also perform sensitivity analysis on the parameter τ (tau) for all BL models in the range of 0.025 to 1, as shown in Table 7, which acts as a robustness check for the BL-FG (Standard) approach. We observe that the BL-FG (Standard) approach still delivers the highest expected returns and Sharpe ratios for all values of tau for all six investment periods compared to all other BL models. This finding proves that the proposed BL-FG approach stands robust and outperforms all other traditional BL approaches for generating views in the prior literature, which is suggestive of the fact that machine learning can be a superior tool for creating BL views compared to econometric approaches and relative BL views better incorporate investor preferences. The results also provide conclusive evidence that investor sentiment prediction can be a valuable tool to guide portfolio decisions.

Table 7Robustness check with variations in tau (τ) for all Black-Litterman portfolios.

		BL-FG (Standard) Expected Return	Sharpe Ratio	BL-PR Expected Return	Sharpe Ratio	BL-EM Expected Return	Sharpe Ratio	BL-SM Expected Return	Sharpe Ratio	BL-Mom Expected Return	Sharpe Ratio
2017	$\tau = 0.025$	26.9%	3.57	19.1%	2.49	12.6%	1.43	8.5%	0.76	27.0%	1.71
	$\tau = 0.05$	27.0%	3.53	19.1%	2.48	12.6%	1.42	8.5%	0.76	26.7%	1.69
	$\tau = 0.50$	28.0%	3.05	19.1%	2.35	12.6%	1.33	8.5%	0.70	23.2%	1.49
	$\tau = 1.00$	29.2%	2.72	19.1%	2.23	12.5%	1.25	8.5%	0.66	21.1%	1.33
2018	$\tau = 0.025$	11.7%	0.49	-8.1%	-0.92	4.6%	0.23	5.0%	0.23	6.3%	0.32
	$\tau = 0.05$	11.8%	0.49	-8.1%	-0.91	4.6%	0.22	5.0%	0.23	6.3%	0.32
	$\tau = 0.50$	13.0%	0.45	-8.2%	-0.86	4.6%	0.21	4.9%	0.21	6.3%	0.30
	$\tau = 1.00$	13.9%	0.42	-8.4%	-0.83	4.5%	0.20	4.7%	0.20	6.3%	0.28
2019	$\tau = 0.025$	23.4%	2.15	19.6%	1.88	9.7%	0.66	17.8%	0.72	15.9%	1.08
	$\tau = 0.05$	23.4%	2.13	19.6%	1.87	9.7%	0.66	17.7%	0.71	15.9%	1.07
	$\tau = 0.50$	23.3%	1.92	19.7%	1.78	9.7%	0.63	16.7%	0.66	16.1%	0.99
	$\tau = 1.00$	23.3%	1.75	19.7%	1.70	9.7%	0.60	15.6%	0.62	16.1%	0.92
2020	$\tau = 0.025$	17.5%	0.69	6.8%	0.29	5.2%	0.23	6.9%	0.25	9.5%	0.31
	$\tau = 0.05$	17.6%	0.69	6.8%	0.29	5.2%	0.22	6.9%	0.25	9.5%	0.30
	$\tau = 0.50$	18.9%	0.62	7.0%	0.28	5.3%	0.21	6.9%	0.24	9.6%	0.29
	$\tau = 1.00$	20.0%	0.57	7.1%	0.27	5.3%	0.20	6.9%	0.23	9.7%	0.28
2021	$\tau = 0.025$	28.9%	2.35	15.9%	1.39	8.9%	0.71	10.2%	0.59	16.5%	1.00
	$\tau = 0.05$	29.0%	2.33	15.9%	1.39	8.9%	0.71	10.2%	0.59	16.4%	1.00
	$\tau = 0.50$	30.5%	2.03	16.0%	1.32	9.0%	0.67	10.1%	0.56	15.6%	0.91
	$\tau = 1.00$	31.7%	1.81	16.1%	1.25	9.1%	0.64	10.0%	0.53	15.1%	0.84
2022	$\tau = 0.025$	14.6%	0.63	-19.3%	-1.18	5.0%	0.18	10.6%	0.42	13.0%	0.50
	$\tau = 0.05$	14.6%	0.63	-19.3%	-1.17	5.0%	0.18	10.6%	0.42	13.0%	0.50
	$\tau = 0.50$	14.8%	0.55	-19.4%	-1.11	4.9%	0.17	10.4%	0.40	12.8%	0.47
	$\tau = 1.00$	15.1%	0.49	-19.6%	-1.08	4.8%	0.15	10.3%	0.38	12.6%	0.44

Note: This table reports the expected return and Sharpe ratio for all BL models with different values of τ (0.025, 0.05, 0.50, 1.00) for the six investment periods. **Legend:** BL-FG = proposed BL approach; BL-PR = BL with views derived via predictive regression; BL-EM = BL with EGARCH-in-mean derived views; BL-SM = BL with sample mean returns as views; BL-Mom = BL with views generated from a momentum strategy.

5. Concluding remarks

The contribution of this letter is twofold. First, we establish that machine learning based approaches are a valuable tool to generate subjective BL views. Combining denoising with deep learning achieves excellent financial predictions, and ensemble learning can best model the relationship between returns and several other predictor variables. Second, we demonstrate how consideration of investor sentiment is vital to classifying assets for constructing relative BL views which can lead to better performance and outperform several other baseline strategies that do not consider investor sentiment. We believe our procedure will meet the requirements of several practitioners for constructing portfolios as it is theoretically grounded on the fact that consideration of market psychology leads to better portfolio management. Future studies can move in the direction of considering better proxies for investor sentiment beyond technical indicators by extracting qualitative sentiment information with techniques such as deep natural language processing and emotion recognition from images via convolutional neural networks to construct more practical relative investor views.

Funding source

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Abbasimehr, H., Paki, R., Bahrini, A., 2023. A novel XGBoost-based featurization approach to forecast renewable energy consumption with deep learning models. *Sustainable Computing: Informatics and Systems* 38, 100863.
- Alhashel, B.S., Almudhaf, F.W., Hansz, J.A., 2018. Can technical analysis generate superior returns in securitized property markets? Evidence from East Asia markets. *Pacific-Basin Finance Journal* 47, 92–108.
- Barua, R., Sharma, A.K., 2022. Dynamic Black Litterman portfolios with views derived via CNN-BiLSTM predictions. *Finance Research Letters* 49, 103111.
- Beach, S.L., Orlov, A.G., 2007. An application of the Black–Litterman model with EGARCH-M-derived views for international portfolio management. *Financial Markets and Portfolio Management* 21, 147–166.
- Bessler, W., Opfer, H., Wolff, D., 2017. Multi-asset portfolio optimization and out-of-sample performance: an evaluation of Black–Litterman, mean-variance, and naïve diversification approaches. *The European Journal of Finance* 23 (1), 1–30.
- Bessler, W., Taushanov, G., Wolff, D., 2021. Optimal asset allocation strategies for international equity portfolios: a comparison of country versus industry optimization. *Journal of International Financial Markets, Institutions and Money* 72, 101343.
- Black, F., Litterman, R., 1990. Asset allocation: combining investor views with market equilibrium. *Goldman Sachs Fixed Income Research* 115 (1), 7–18.
- Black, F., Litterman, R., 1992. Global portfolio optimization. *Financial Analysts Journal* 48 (5), 28–43.
- Cao, J., Li, Z., Li, J., 2019. Financial time series forecasting model based on CEEMDAN and LSTM. *Physica A: Statistical mechanics and its applications* 519, 127–139.
- Cengiz, C.B., Herwartz, H., 2011. Modeling stock index returns by means of partial least-squares methods: An out-of-sample analysis for three stock markets. *Applied Stochastic Models in Business and Industry* 27 (3), 253–266.
- Chen, T., Guestrin, C., 2016. Xgboost: A scalable tree boosting system. In: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794.
- Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint*.
- Creamer, G.G., 2015. Can a corporate network and news sentiment improve portfolio optimization using the Black–Litterman model? *Quantitative Finance* 15 (8), 1405–1416.
- Day, M.Y., Ni, Y., Huang, P., 2019. Trading as sharp movements in oil prices and technical trading signals emitted with big data concerns. *Physica A: Statistical Mechanics and its Applications* 525, 349–372.
- Dezhkam, A., Manzuri, M.T., 2023. Forecasting stock market for an efficient portfolio by combining XGBoost and Hilbert–Huang transform. *Engineering Applications of Artificial Intelligence* 118, 105626.
- Duqi, A., Franci, L., Torluccio, G., 2014. The Black–Litterman model: the definition of views based on volatility forecasts. *Applied Financial Economics* 24 (19), 1285–1296.
- Fabozzi, F.J., Focardi, S.M., Kolm, P.N., 2006. Incorporating trading strategies in the Black–Litterman framework. *The Journal of Trading* 1 (2), 28–37.
- Fang, J., Qin, Y., Jacobsen, B., 2014. Technical market indicators: An overview. *Journal of behavioral and experimental finance* 4, 25–56.
- Feng, Z.K., Huang, Q.Q., Niu, W.J., Yang, T., Wang, J.Y., Wen, S.P., 2022. Multi-step-ahead solar output time series prediction with gate recurrent unit neural network using data decomposition and cooperation search algorithm. *Energy* 261, 125217.
- Fernandes, J.L.B., Ornelas, J.R.H., Cusicanqui, O.A.M., 2012. Combining equilibrium, resampling, and analyst's views in portfolio optimization. *Journal of Banking & Finance* 36 (5), 1354–1361.
- Fernández, M.F., Henry, Ó., Pybis, S., Stamatogiannis, M.P., 2023. Can we forecast better in periods of low uncertainty? The role of technical indicators. *Journal of Empirical Finance* 71, 1–12.
- Fischer, T., Krauss, C., 2018. Deep learning with long short-term memory networks for financial market predictions. *European journal of operational research* 270 (2), 654–669.
- Gao, B., Yang, C., 2017. Forecasting stock index futures returns with mixed-frequency sentiment. *International Review of Economics & Finance* 49, 69–83.
- Gu, S., Kelly, B., Xiu, D., 2020. Empirical asset pricing via machine learning. *The Review of Financial Studies* 33 (5), 2223–2273.
- Gupta, K., Chatterjee, N., 2021. Stocks Recommendation from Large Datasets Using Important Company and Economic Indicators. *Asia-Pacific Financial Markets* 28 (4), 667–689.
- Han, Y., Li, J., 2023. The impact of global economic policy uncertainty on portfolio optimization: A Black–Litterman approach. *International Review of Financial Analysis* 86, 102476.
- Hanauer, M.X., Kalsbach, T., 2023. Machine learning and the cross-section of emerging market stock returns. *Emerging Markets Review* 55, 101022.
- He, G., Litterman, R., 2002. The Intuition Behind Black–Litterman Model Portfolios. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.334304>.
- Huddleston, D., Liu, F., Stentoft, L., 2023. Intraday market predictability: A machine learning approach. *Journal of Financial Econometrics* 21 (2), 485–527.
- Idzorek, T., 2007. A step-by-step guide to the Black–Litterman model: Incorporating user-specified confidence levels. *Forecasting Expected Returns in the Financial Markets*. Academic Press, pp. 17–38.
- Kara, M., Ulucan, A., Atici, K.B., 2019. A hybrid approach for generating investor views in Black–Litterman model. *Expert Systems with Applications* 128, 256–270.
- Khayamim, A., Mirzazadeh, A., Naderi, B., 2018. Portfolio rebalancing with respect to market psychology in a fuzzy environment: a case study in Tehran Stock Exchange. *Applied Soft Computing* 64, 244–259.
- Kolm, P.N., Ritter, G., Simonian, J., 2021. Black–litterman and beyond: The bayesian paradigm in investment management. *The Journal of Portfolio Management* 47 (5), 91–113.
- Krauss, C., Do, X.A., Huck, N., 2017. Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research* 259 (2), 689–702.
- Leippold, M., Wang, Q., Zhou, W., 2022. Machine learning in the Chinese stock market. *Journal of Financial Economics* 145 (2), 64–82.
- Lin, W.H., Teng, H.W., & Yang, C.C. (2021). A heteroskedastic Black–Litterman portfolio optimization model with views derived from a predictive regression. In *Handbook of Financial Econometrics, Mathematics, Statistics, and Machine Learning* (pp. 563–581).

- Lin, Y., Lin, Z., Liao, Y., Li, Y., Xu, J., Yan, Y., 2022. Forecasting the realized volatility of stock price index: A hybrid model integrating CEEMDAN and LSTM. *Expert Systems with Applications* 206, 117736.
- Lu, H., Ma, X., 2020. Hybrid decision tree-based machine learning models for short-term water quality prediction. *Chemosphere* 249, 126169.
- Ma, F., Lu, X., Liu, J., Huang, D., 2022. Macroeconomic attention and stock market return predictability. *Journal of International Financial Markets, Institutions and Money* 79, 101603.
- Ma, Y., Wang, Y., Wang, W., Zhang, C., 2023. Prediction-based mean–variance portfolios with risk budgeting based on neural networks. *Expert Systems with Applications*, 120638.
- Mourtas, S.D., Katsikis, V.N., 2022. Exploiting the Black–Litterman framework through error-correction neural networks. *Neurocomputing* 498, 43–58.
- Mullainathan, S., Spiess, J., 2017. Machine learning: an applied econometric approach. *Journal of Economic Perspectives* 31 (2), 87–106.
- Neely, C.J., Rapach, D.E., Tu, J., Zhou, G., 2014. Forecasting the equity risk premium: the role of technical indicators. *Management science* 60 (7), 1772–1791.
- Oikonomou, I., Platanakis, E., Sutcliffe, C., 2018. Socially responsible investment portfolios: does the optimization process matter? *The British Accounting Review* 50 (4), 379–401.
- Pyo, S., Lee, J., 2018. Exploiting the low-risk anomaly using machine learning to enhance the Black–Litterman framework: Evidence from South Korea. *Pacific-Basin Finance Journal* 51, 1–12.
- Rezaei, H., Faaljou, H., Mansourfar, G., 2021. Intelligent asset allocation using predictions of deep frequency decomposition. *Expert Systems with Applications* 186, 115715.
- Shefrin, H., 2002. *Beyond Greed and fear: Understanding behavioral Finance and the Psychology of Investing*. Oxford University Press on Demand.
- Silva, T., Pinheiro, P.R., Poggi, M., 2017. A more human-like portfolio optimization approach. *European Journal of Operational Research* 256 (1), 252–260.
- Stein, T., 2022. Forecasting the equity premium with frequency-decomposed technical indicators. *International Journal of Forecasting*.
- Tang, M.L., Wu, F.Y., Hung, M.C., 2021. Multi-asset allocation of exchange traded funds: Application of Black–Litterman model. *Investment Analysts Journal* 50 (4), 273–293.
- Torres, M.E., Colominas, M.A., Schlotthauer, G., Flandrin, P., 2011. A complete ensemble empirical mode decomposition with adaptive noise. In: 2011 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp. 4144–4147.
- Walters, J. (2014). *The Black-Litterman Model in Detail*. SSRN. <https://doi.org/10.2139/ssrn.1314585>.
- Wang, J., Cao, J., Yuan, S., Cheng, M., 2021. Short-term forecasting of natural gas prices by using a novel hybrid method based on a combination of the CEEMDAN-SE and the PSO-ALS-optimized GRU network. *Energy* 233, 121082.
- Wu, W., Chen, J., Xu, L., He, Q., Tindall, M.L., 2019. A statistical learning approach for stock selection in the Chinese stock market. *Financial Innovation* 5 (1), 1–18.
- Yang, C., Gao, B., 2014. The term structure of sentiment effect in stock index futures market. *The North American Journal of Economics and Finance* 30, 171–182.
- Yang, C., Zhou, L., 2015. Investor trading behavior, investor sentiment and asset prices. *The North American Journal of Economics and Finance* 34, 42–62.
- Yang, C., Zhou, L., 2016. Individual stock crowded trades, individual stock investor sentiment and excess returns. *The North American Journal of Economics and Finance* 38, 39–53.